

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch : AI& Allied Engg

Semester: III

Subject Code & Name: Digital Logic & Signal Processing (BTESC305 / BTAIES305)

Max Marks: 60

Date:11-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is difference between Universal Gates and Basic Gates? Also implement AND gate using NOR gate only.	CO1	
B) What is difference between weighted and non weighted code? Give examples and also write a short note on Grey Code.	CO1	
C) Explain 1's Complement and 2's Complement with example.	CO1	
Q.2 Solve Any Two of the following.		12
A) What is System? Classify and explain it briefly.	CO2	
B) Explain following in terms of operation on signals 1) Time Scaling 2) Time Reversal 3) Time Shifting	CO2	
C) Find whether the following system is linear or not: $y(n) = x(-n+2)$	CO2	
Q. 3 Solve Any Two of the following.		12
A) What is difference between DFT and FFT? Explain the properties of DFT.	CO3	
B) Compute linear convolution of given sequence: $h(n) = \{1, 1, 1, 1\}$ and $x(n) = \{1, 1, 1, 1\}$	CO3	
C) Compute DFT of four point sequence: $x(n) = \{1, 0, 1, 0\}$	CO3	
Q.4 Solve Any Two of the following.		12
A) What is ROC? Explain its properties.	CO4	
B) Find Z transform of signals, $x(n) = n a^n u(n)$	CO4	
C) Find Inverse Z transform of $X(Z) = Z / (3Z^2 - 4Z + 1)$	CO4	
Q. 5 Solve Any Two of the following.		12
A) Explain analogy between correlation and convolution.	CO5	
B) What is ESD? Also explain properties of ESD with example.	CO5	
C) What is correlation? Also explain properties of cross correlation with example.	CO5	

***** End *****